

Maximal multipoint Seshadri constants in positive characteristic

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Abstract

Let L be an ample line bundle on a smooth integral projective variety X over an algebraically closed field of positive characteristic, with $\dim X = n \geq 3$. We prove that there is a threshold $r_0 = r_0(X, L)$ such that for every integer $r \geq r_0$, the ordinary multipoint Seshadri constant at the geometric generic tuple of r points equals the volume bound $(L^n/r)^{1/n}$. The same conclusion holds in dimension two. This establishes the positive-characteristic form of the qualitative Nagata–Biran–Szemberg assertion at geometric generic tuples.

1 Introduction

Multipoint Seshadri constants measure how much positivity an ample line bundle retains after imposing simultaneous conditions at several points. Their volume bound depends only on the top self-intersection of the line bundle and the number of points. The question is whether, for sufficiently many general points, any curve can impose a stricter bound. We prove that no such curve exists at a geometric generic tuple in positive characteristic.

Let k be an algebraically closed field of characteristic $p > 0$, let X/k be a smooth integral projective variety of dimension n , and let L be an ample line bundle. For $r \geq 1$, write

$$U_r = \{(x_1, \dots, x_r) \in X^r : x_i \neq x_j \text{ for } i \neq j\}$$

for the open configuration space of ordered distinct points. Put $K_r = \overline{k(U_r)}$, and let $p_1, \dots, p_r \in X(K_r)$ be the points specified by its geometric generic point. Define

$$\varepsilon_r(X, L) = \inf_C \frac{L_{K_r} \cdot C}{\sum_{i=1}^r \text{mult}_{p_i} C}, \quad (1.1)$$

where C ranges over integral curves in X_{K_r} meeting at least one p_i . Multiplicity at a point outside C is zero. Throughout, we use this ordinary, curve-defined Seshadri constant.

Theorem 1.1. *Let X/k be a smooth integral projective variety over an algebraically closed field of characteristic $p > 0$, with $\dim X = n \geq 3$, and let L be ample. There is an integer $r_0 = r_0(X, L)$ such that*

$$\varepsilon_r(X, L) = \left(\frac{L^n}{r} \right)^{1/n} \quad \text{for every integer } r \geq r_0.$$

The same conclusion holds when $n = 2$; see Remark 6.3.

The geometric generic formulation includes countable algebraically closed fields. It also gives a precise meaning to general position without choosing points in an infinite intersection

of open subsets. The equality is exact for each sufficiently large point count. On the blow-up of the geometric generic points, it says that

$$\pi^* L_{K_r} - (L^n/r)^{1/n} \sum_{i=1}^r E_i$$

is a nef real divisor class with top self-intersection zero.

1.1 History and the characteristic issue

Nagata’s work on Hilbert’s fourteenth problem led to his conjecture on the degrees of plane curves with assigned multiplicities at general points, and to its proof when the number of points is a square at least 16 [9]. The qualitative Nagata–Biran–Szemberg assertion extends eventual maximality to polarized varieties. Ro   and Ross formulate it in arbitrary dimension over uncountable algebraically closed fields and distinguish the ordinary Seshadri constant from its higher-dimensional cycle analogues [14, Conjecture 1.3 and Remark 1.2]. Theorem 1.1 proves its positive-characteristic form, with general position expressed by the geometric generic tuple, in the stated dimensions.

Several preceding results explain the volume scale of this question. Biran’s packing stability theorem gives a symplectic analogue on closed four-manifolds with rational symplectic class [1]. K  hle obtains asymptotically optimal lower bounds for multipoint Seshadri constants on complex projective varieties [6]. Product inequalities compare point counts on a variety with those on projective space [14, Theorem 1.1 and Remark 1.2]. These are distinct from exact algebraic maximality for every sufficiently large integer r .

Our proof adapts the graded-support construction of the companion *Maximal Multipoint Seshadri Constants in Higher Dimensions* [11, Sections 2–6], which treats complex varieties. Its two-coordinate reshaping follows the nodal exponent bounds and primitive lattice directions in the surface companion [12, Propositions 3.2 and 3.4 and Lemma 4.3]. We prove every required statement here. The passage to positive characteristic requires an algebraic replacement for analytic limits and a nodal equation that retains two distinct branches in characteristic two. Formal substitutions over $k[[s]]$ supply the first; the curve $y^2 + xy - x^3 = 0$ supplies the second.

The relation between positivity and asymptotic jet separation is classical [2, Theorem 6.4]. The use of leading monomials belongs to the valuation approach to linear series [5, 7, 10]; Ito gives finite-degree comparisons between multipoint jet separation and initial monomial spaces [4, Proposition 5.7 and Theorem 5.8]. Park and Shin give characteristic-independent Okounkov-body descriptions of one-point Seshadri constants for complete series of nef and big divisors [13, Theorem 1.2(1)]. Monomial interpolation also underlies Dumnicki’s diagram-cutting method [3, Proposition 13 and Theorem 14]. The statements below isolate two useful forms of these ideas: formal transfer of arbitrary finite jets, and reshaping of a graded family with prescribed real simplex intercepts. The latter keeps both the exact cardinality in every degree and multiplication between degrees. A short counting argument then turns full asymptotic density into exact filling of compact subsets of the interior.

1.2 Proof strategy

Put $V = L^n$ and, for a fixed point count r , put $w = (V/r)^{1/n}$. For a line bundle A at a smooth point x , its jets through order m form $A_x/\mathfrak{m}_x^{m+1}A_x$. Separating those jets at a tuple means that sections surject onto the direct sum of these spaces. Such separation by $H^0(X, L^{\otimes N})$ gives the curve inequality

$$NL \cdot C \geq m \sum_i \text{mult}_{p_i} C.$$

It therefore suffices to obtain jets through order $\lfloor N\theta \rfloor$ for every fixed $0 < \theta < w$ and all sufficiently large N .

We encode sections by finite sets $S_N \subset \mathbb{Z}^n$ of leading exponents, with the multiplication property

$$S_N + S_M \subseteq S_{N+M}.$$

Their monomial spaces on $\mathbb{T}^n = (\mathbb{G}_{m,k})^n$ are $M(S_N) = \text{span}_k\{z^\alpha : \alpha \in S_N\}$. Section 2 shows how jet separation for such monomials can be transferred to the original functions or sections. The transfer is a finite-degree rank argument. Its auxiliary weights may depend on N , while the orders defining the graded sets remain fixed.

For positive intercepts $a_1, \dots, a_n$, write

$$\Delta(a_1, \dots, a_n) = \left\{ x \in \mathbb{R}_{\geq 0}^n : \sum_i x_i/a_i \leq 1 \right\}.$$

A complete-intersection flag in Section 3 gives

$$S_N \subseteq N\Delta(1/D, \dots, 1/D, D^{n-1}V), \quad \#S_N = h^0(X, L^{\otimes N}),$$

where DL is very ample. The simplex has volume $V/n!$, equal to the leading coefficient of the section count.

The main geometric step, Section 4, replaces two intercepts A, B by $w, AB/w$ for any sufficiently small prescribed real $w > 0$. It preserves cardinalities, the graded property, and the implication from jet separation after the replacement to jet separation before it. For a suitable leading-term order, a node bounds the exponents by a quadrilateral of the same area. Integral changes of coordinates and successive compressions, which move interval slices to start at zero, turn this quadrilateral into the desired triangle. The other coordinates serve only as labels.

For large r , repeating this step yields graded supports F_N inside

$$NP_*, \quad P_* = \Delta(w, \dots, w, rw), \quad \#F_N = \frac{V}{n!}N^n + o(N^n).$$

Section 5 proves that every lattice point in a fixed compact interior region belongs to F_N in large degree. Indeed, each such point has on the order of N^n decompositions into two nearly equal degrees, whereas the missing points exclude only $o(N^n)$ of them. The long final intercept of P_* then permits polynomial interpolation at r points differing only in their last coordinate. This interpolation uses the Chinese remainder theorem and is valid in every characteristic.

Each transferred jet-separation statement holds on a nonempty open subset of U_r , hence at its geometric generic point. Section 6 converts these statements into the required lower bound for every curve and proves the matching volume upper bound on the blow-up.

2 Leading terms and transfer of jets

We first turn spaces of sections into finite sets of exponents. The essential point is that jet separation by the resulting monomials implies jet separation by the original sections. We will also make two elementary changes to the exponent sets while retaining this implication. These operations follow the support method of [11, Section 2]; the formal transfer below makes the method available in arbitrary characteristic.

Throughout, k is algebraically closed. The arguments of this section work in every characteristic. For a smooth d -dimensional variety Y , a line bundle A , and $y \in Y(k)$, the jets through order $m \geq 0$ are

$$J_y^m(A) = A_y/\mathfrak{m}_y^{m+1}A_y.$$

A finite-dimensional space $W \subset H^0(Y, A)$ separates these jets at $y_1, \dots, y_r$ if the evaluation map $W \rightarrow \bigoplus_i J_{y_i}^m(A)$ is surjective. The locus of such ordered distinct tuples is open in Y^r : the jet spaces are the fibers of the locally free sheaf of principal parts of A , of rank $\binom{d+m}{d}$, and surjectivity is a rank condition. The description of principal parts by the m th infinitesimal neighborhood of the diagonal also shows compatibility with field extension. Local coordinates for a smooth variety identify its fibers with truncated polynomial rings. For the diagonal description and base change, see [15, Tags 0H90 and 0H8Z]. Local freeness follows as well from the filtration whose graded pieces are $A \otimes \text{Sym}^j \Omega_Y^1$, $0 \leq j \leq m$.

Put $\mathbb{T}^d = (\mathbb{G}_{m,k})^d$. For a finite set $S \subset \mathbb{Z}^d$, write

$$M(S) = \text{span}_k \{x^\alpha : \alpha \in S\} \subset k[x_1^{\pm 1}, \dots, x_d^{\pm 1}].$$

Here and below, auxiliary existence statements about points on a torus refer to k -points.

Definition 2.1. A family of finite sets $(S_N)_{N \geq 1}$ in $\mathbb{Z}^d$ is *graded* if

$$S_N + S_{N'} \subset S_{N+N'} \quad (N, N' \geq 1).$$

A transformation $(S_N) \mapsto (T_N)$ has *backward jet transfer* if, for every $N, r \geq 1$ and $m \geq 0$, separation of jets through order m by $M(T_N)$ at some r distinct torus points implies the same assertion for $M(S_N)$, possibly at a different tuple.

2.1 Least exponents

A monomial order on $\mathbb{Z}_{\geq 0}^h$ means an addition-compatible total well-order. For a nonzero formal series $f \in k[[z_1, \dots, z_h]]$, let $\nu(f)$ be the least exponent occurring in f , for a fixed such order. In particular,

$$\nu(fg) = \nu(f) + \nu(g). \quad (2.1)$$

Indeed, the product of the two least terms is the unique least term in the product.

Lemma 2.2. For a finite-dimensional subspace $W \subset k[[z_1, \dots, z_h]]$, the set $\{\nu(f) : 0 \neq f \in W\}$ has cardinality $\dim_k W$. More generally, suppose finite-dimensional spaces $W_{N,\gamma}$, with only finitely many nonzero spaces in each degree, are indexed by $N \geq 1$ and $\gamma \in \mathbb{Z}^{d-h}$ and satisfy

$$W_{N,\gamma} W_{N',\gamma'} \subset W_{N+N',\gamma+\gamma'}.$$

Then recording the pairs $(\nu(f), \gamma)$ for $0 \neq f \in W_{N,\gamma}$ gives a graded family, with cardinality $\sum_\gamma \dim_k W_{N,\gamma}$ in degree N .

Proof. Choose the smallest exponent occurring as $\nu(f)$ for a nonzero member of W , and choose a member whose coefficient there is 1. The kernel of that coefficient functional has codimension one. Repeating in this kernel gives a basis with distinct least exponents. Every nonzero linear combination has as its least exponent the smallest least exponent among the basis vectors with nonzero coefficients. This proves the first assertion; the second follows label by label and from (2.1). $\square$

For the transfer of jets we need to realize finitely many chosen least terms by a single positive weight vector. The weights may depend on the degree; the monomial order defining the graded family will remain fixed.

Lemma 2.3. Let $f_1, \dots, f_q$ be nonzero formal series, and let $e_j = \nu(f_j)$. There is a vector $\lambda \in \mathbb{Z}_{>0}^h$ for which e_j is the unique exponent of least λ -weight in f_j , simultaneously for all j , in either of the following cases:

- (i) *the order is lexicographic;*
- (ii) *$h = 2$ and, for a fixed real $t > 0$, the order compares $(\alpha + t\beta, \beta)$ lexicographically.*

Proof. In the lexicographic case set $\lambda_h = 1$, and successively choose

$$\lambda_i > \max_j \sum_{a>i} \lambda_a (e_j)_a \quad (i = h-1, \dots, 1).$$

If an exponent is lexicographically greater than e_j , its gain at the first differing coordinate outweighs every possible loss in the later coordinates.

For (ii), the indicated order is a monomial order because every bounded weight range contains only finitely many nonnegative integral exponents. Choose a real bound larger than the weights of all the e_j . Among the finitely many exponents below that bound, replacing t by a sufficiently close rational $t' > t$ preserves all strict comparisons with the e_j and resolves equal-weight comparisons by increasing β . Exponents above the bound cannot interfere: their weights only increase, while the new weights of the e_j stay below the bound. Clearing the denominator of $(1, t')$ gives the required positive integral weights. $\square$

2.2 Formal specialization of jet matrices

The next lemma supplies the passage from formal leading terms to actual points. Ito's weighted degeneration argument gives such a jet-rank comparison over $\mathbb{C}$ [4, Proposition 5.7]. Here substitution in truncated rings supplies a formal version in arbitrary characteristic, without differentiation or factorial denominators.

Lemma 2.4 (Formal jet transfer). *Let $1 \leq h \leq d$, let Y be a smooth integral h -dimensional k -variety, and choose $y \in Y(k)$ with formal coordinates $z_1, \dots, z_h$. Let $f_1, \dots, f_q$ be regular functions on a common neighborhood of y , and let $\gamma_j \in \mathbb{Z}^{d-h}$. Suppose the expansion of f_j has a term $c_j z^{e_j}$, with $c_j \neq 0$, which is uniquely lowest for one common weight vector $\lambda \in \mathbb{Z}_{>0}^h$.*

If the span of the monomials $Z^{e_j} u^{\gamma_j}$ separates jets through order m at r distinct points of $\mathbb{T}^h \times \mathbb{T}^{d-h}$, then the span of $f_j u^{\gamma_j}$ separates the same jets at r distinct points in the product of that neighborhood with $\mathbb{T}^{d-h}$. The points may be chosen in any prescribed dense open subset of this product. The statement also holds for sections written in a local frame of a line bundle.

Proof. Let (Q_i, U_i) , $1 \leq i \leq r$, be a tuple at which the monomials separate the stated jets. Shrink the neighborhood of y to an affine one, denoted Y° , on which all the functions are defined. Put

$$R = k[[s]], \quad F = k((s)), \quad B = R[\delta_1, \dots, \delta_d] / (\delta_1, \dots, \delta_d)^{m+1},$$

and write $\delta = (\delta', \delta'')$ according to the two factors. For each i , substitute

$$z_a = s^{\lambda_a} (Q_{i,a} + \delta'_a), \quad u_b = U_{i,b} + \delta''_b. \quad (2.2)$$

The first substitution is defined on the completed local ring, since B is s -adically complete and every λ_a is positive. The second is defined on Laurent polynomials, since each $U_{i,b}$ is nonzero. In particular, these substitutions define ring maps from the coordinate ring of $Y^\circ \times \mathbb{T}^{d-h}$ to B .

Use the classes of δ^α , $|\alpha| \leq m$, as an R -basis of B . The images of $f_j u^{\gamma_j}$ under all r substitutions give the j th column of a matrix with $r \binom{d+m}{d}$ rows. Divide this column by $c_j s^{\lambda \cdot e_j}$. These normalized columns still belong to $B^{\oplus r}$, and their reductions modulo s are precisely

the jets of $Z^{e_j} u^{\gamma_j}$ at (Q_i, U_i) in translated coordinates. The assumed surjectivity supplies a maximal minor nonzero modulo s . The original columns therefore span $B[1/s]^{\oplus r}$ over F .

We now interpret this rank statement as jet separation on a variety. Setting $\delta = 0$ in (2.2) gives actual F -points P_i of $Y^\circ \times \mathbb{T}^{d-h}$. After inverting s , the corresponding ring map factors through the order- m jet algebra of $(Y^\circ \times \mathbb{T}^{d-h})_F$ at P_i : elements nonvanishing at P_i map to units, and its maximal ideal maps into (δ) . Both this jet algebra and $B[1/s]$ have dimension $\binom{d+m}{d}$ over F . Surjectivity onto the direct sum of the latter spaces forces each induced map from a jet algebra to be an isomorphism. It follows that the original functions separate the order- m jets at the P_i themselves. These points are distinct, since two repeated centers would make two target blocks factor through one jet algebra, contradicting surjectivity.

Thus the open jet-separation locus has an F -point and is nonempty. The configuration space is irreducible, so this locus meets the configuration space of any prescribed dense open subset. A nonempty open subset of a variety over the algebraically closed field k has a k -point. This proves the assertion. A local frame reduces the line-bundle version to the one just proved. $\square$

Together, Lemmas 2.2–2.4 allow us to replace finite-dimensional spaces by their least exponents, using either of the two orders above, while preserving dimension and transferring jet separation back to the original spaces. Labels record the exponents of variables that are left unchanged.

2.3 Two elementary transformations

Lemma 2.5. *For $A \in \mathrm{GL}_d(\mathbb{Z})$, the transformation $S_N \mapsto AS_N$ preserves cardinalities and gradedness, and has backward jet transfer.*

Proof. An invertible integral linear map preserves cardinalities and addition. Its action on characters is induced by a torus automorphism, which identifies the corresponding jet evaluations. $\square$

The other transformation uses orders of vanishing to place each row in an interval starting at zero. The resulting exponents need not be consecutive.

Lemma 2.6 (Coordinate compression). *Let (S_N) be a graded family in $\mathbb{Z}^d$, and select one coordinate, written z . For each degree N and each label $\gamma \in \mathbb{Z}^{d-1}$, form its row space*

$$W_{N,\gamma} = \mathrm{span}_k\{z^a : (a, \gamma) \in S_N\}.$$

Replace this row by the set of orders at $z = 1$ of nonzero elements of $W_{N,\gamma}$, retaining the label γ . The resulting family preserves cardinalities and gradedness, and has backward jet transfer. If the original row lies in a real interval of length b , every new exponent lies in $[0, b]$.

Proof. Expand the row spaces in the parameter $z - 1$. Lemma 2.2 proves the cardinality and gradedness assertions, because multiplication adds both degrees and labels. Lemma 2.4, with $Y = \mathbb{G}_m$, $y = 1$, and weight 1, proves backward jet transfer.

For a nonempty row let a_- and a_+ be its smallest and largest exponents. Multiplication by z^{-a_-} preserves orders at 1 and turns each member into a polynomial of degree at most $a_+ - a_- \leq b$. Its order of vanishing is at most its degree, proving the bound. $\square$

Consequently, suppose a family is bounded by dilates of a polygon. Replacing each slice of that polygon parallel to the compressed coordinate by an interval of the same length starting at zero gives a new region whose dilates contain the compressed family. This applies equally to real slice endpoints and to every dilation of the polygon. In characteristic p , a compressed row need not contain all intermediate integers: for example, $\mathrm{span}_k\{1, z^p\}$ has orders 0 and p at 1. Only the interval bound of Lemma 2.6 will be used.

3 An initial simplex from a flag

We now apply the leading-term construction to the section ring of L . A complete-intersection flag gives a particularly simple bound on the exponents: a simplex whose volume equals the leading coefficient of the Hilbert polynomial. We give the complete argument for the flag construction of [11, Section 3].

For positive real numbers $a_1, \dots, a_d$, write

$$\Delta(a_1, \dots, a_d) = \left\{ x \in \mathbb{R}_{\geq 0}^d : \sum_{i=1}^d \frac{x_i}{a_i} \leq 1 \right\}.$$

Its volume is $\prod_i a_i / d!$. We use additive notation NL for $L^{\otimes N}$.

Proposition 3.1. *Let X be a smooth integral projective n -dimensional variety over an algebraically closed field, with $n \geq 2$, and let L be ample. Put $V = L^n$, and choose an integer $D \geq 1$ such that DL is very ample. There is a graded family of finite sets $S_N \subset \mathbb{Z}_{\geq 0}^n$ such that*

$$S_N \subset NP_0, \quad P_0 = \Delta(1/D, \dots, 1/D, D^{n-1}V), \quad (3.1)$$

$$\#S_N = h^0(X, NL) = \frac{V}{n!} N^n + o(N^n). \quad (3.2)$$

For every $N, r \geq 1$ and $m \geq 0$, if $M(S_N)$ separates jets through order m at some r distinct torus points, then $H^0(X, NL)$ separates the same jets at some r distinct points of X .

Proof. Bertini's theorem for hyperplane sections, valid in every characteristic for a projective embedding over an algebraically closed field, gives sections $\sigma_1, \dots, \sigma_{n-1} \in H^0(X, DL)$ whose successive zero loci form a smooth integral flag

$$X = X_0 \supset X_1 \supset \dots \supset X_{n-1} = C_0.$$

See [15, Tags 0FD6 and 0G4G] for the smoothness and integrality assertions. At each step the variety being cut has dimension at least two. Choose $x \in C_0(k)$ and a local frame e of L near x . The functions $z_i = \sigma_i/e^D$, $i < n$, form part of a regular system of parameters at x ; complete them by z_n . These parameters identify $\hat{\mathcal{O}}_{X,x}$ with $k[[z_1, \dots, z_n]]$.

Expand every section of NL using the frame e^N , and let S_N be the set of lexicographically least exponents of the nonzero expansions. The expansion map is injective: a section with zero germ is zero by integrality, and the local ring injects into its completion. Lemma 2.2 therefore gives $\#S_N = h^0(X, NL)$ and the graded property. The Hilbert polynomial of the ample line bundle gives the last equality in (3.2). A basis with distinct least exponents, Lemma 2.3(i), and Lemma 2.4 with $h = n$ prove the jet-transfer assertion.

It remains to prove the simplex bound. Let $\alpha = (\alpha_1, \dots, \alpha_n) \in S_N$, coming from a nonzero section s . We successively divide by the equations of the flag and restrict to the next member. The details matter because the lexicographic order first measures vanishing along an entire divisor, not just at the chosen point.

Suppose we have reached a nonzero section on X_{i-1} with least exponent $(\alpha_i, \dots, \alpha_n)$ in its formal expansion at x . If $\alpha_i > 0$, restriction of this section to X_i has zero formal expansion. Injectivity into the completed local ring, followed by integrality of X_i , shows that this restriction is identically zero. The exact sequence for the effective Cartier divisor $X_i \subset X_{i-1}$ therefore allows division by $\sigma_i|_{X_{i-1}}$. Repeating this argument α_i times, and then restricting to X_i , gives a nonzero section whose least exponent is $(\alpha_{i+1}, \dots, \alpha_n)$. Nonvanishing follows from the nonzero coefficient of the original least term. Each division subtracts DL from the line bundle.

After these steps, the resulting nonzero section on C_0 belongs to

$$\left(N - D \sum_{i < n} \alpha_i\right) L|_{C_0}$$

and has order exactly α_n at x . Since $L \cdot C_0 = D^{n-1}V$, comparison with the degree of its zero divisor gives

$$\alpha_n \leq \left(N - D \sum_{i < n} \alpha_i\right) D^{n-1}V.$$

Equivalently, $D \sum_{i < n} \alpha_i + \alpha_n / (D^{n-1}V) \leq N$, which is (3.1). $\square$

The intercepts of P_0 have product V . The next section changes them, two at a time, without changing this product, the numbers of exponents, or backward jet transfer.

4 Reshaping two intercepts

We now change two intercepts of a simplex while preserving their product. The support sets need not fill the simplex: the construction applies to any graded family inside it. Its algebraic step uses the two branches of a node; its remaining steps are lattice changes and coordinate compressions. This follows the planar construction of OpenAI [11, Section 4, Proposition 4.1], with a formal node whose branches remain distinct in characteristic two. The nodal bound, compressions, and primitive-direction choice also appear in OpenAI [12, Propositions 3.2 and 3.4, Lemma 4.3]; we give all the arguments needed here.

Proposition 4.1 (Two-intercept reshaping). *Let $n \geq 2$, let $A, B, a_3, \dots, a_n > 0$, and let $(S_N)_{N \geq 1}$ be a graded family of finite subsets of $\mathbb{Z}^n$ satisfying*

$$S_N \subset N\Delta(A, B, a_3, \dots, a_n).$$

Put $H = AB$ and $d = \max\{3/A, 2/B\}$. For every $w > 0$ with $dw < 1/16$, there is a graded family $(S'_N)_{N \geq 1}$ such that

$$S'_N \subset N\Delta(w, H/w, a_3, \dots, a_n), \quad \#S'_N = \#S_N \quad (N \geq 1).$$

The transformation from S_N to S'_N has backward jet transfer in every degree. For $n = 2$, the lists $a_3, \dots, a_n$ are empty.

We first prove the nodal bound and then choose the lattice direction that produces the prescribed intercept w . Throughout this section, H and d have the meanings above, and we put

$$W = dH = \max\{2A, 3B\}, \quad c = 1/d, \quad T = d^2H = \max\{9B/A, 4A/B\} \geq 6. \quad (4.1)$$

In particular, $cW = H$.

4.1 The two branches of a node

For $t > 0$, order pairs $(\alpha, \beta) \in \mathbb{Z}_{\geq 0}^2$ by the lexicographic order of $(\alpha + t\beta, \beta)$, as in Lemma 2.3. We denote the least exponent in this order by ν_t .

Lemma 4.2 (Nodal support bound). *There are formal coordinates ξ, η at the origin of $\mathbb{A}_k^2$ with*

$$y^2 + xy - x^3 = \xi\eta$$

for which the following assertion holds. Given $t > 0$, set

$$a = \frac{Wt}{1+t}, \quad b = \frac{W}{1+t}, \quad Q = \text{conv}\{(0,0), (a,0), (c,c), (0,b)\}.$$

If $\kappa \geq 0$ and $0 \neq f \in k[x,y]$ has all its exponents in $\kappa\Delta(A,B)$, then its expansion in these coordinates satisfies $\nu_t(f) \in \kappa Q$.

Proof. Write $g = y^2 + xy - x^3$. The parametrization

$$x = v(v+1), \quad y = v^2(v+1) = vx \tag{4.2}$$

identifies the complement of $x = 0$ in $g = 0$ with $\text{Spec } k[v, 1/(v(v+1))]$: the inverse is $v = y/x$. Consequently, a polynomial vanishes under (4.2) if and only if it is divisible by g . Indeed, vanishing implies membership in (g) after inverting x , and g is relatively prime to x in the unique factorization domain $k[x,y]$.

The polynomial $Z^2 + Z - x$ has simple roots 0 and -1 modulo x . Hensel lifting gives roots $r_1, r_2 \in k[[x]]$ with these respective constant terms. Set

$$\eta = y - xr_1(x), \quad \xi = y - xr_2(x).$$

Their linear terms are y and $y + x$, so they are formal coordinates, and $\xi\eta = g$. These assertions also hold in characteristic two: the roots 0 and $-1 = 1$ remain distinct and the linear terms remain independent. Along (4.2) near $v = 0$, uniqueness of the lifted root gives $r_1(x(v)) = v$, so $\eta = 0$ and ξ has order one in v . Near $v = -1$, one similarly has $\xi = 0$ and η has order one in $v + 1$.

Factor $f = g^e f_0$ with $e \geq 0$ and $g \nmid f_0$. Give x, y the positive weights $1/A, 1/B$, and write $\deg_{A,B}$ for the maximum weight of a polynomial's monomials. This degree is additive on products, because their highest weight forms multiply nontrivially. Moreover,

$$\deg_{A,B}(g) = d, \quad \deg_{A,B}(f_0) \leq \kappa - ed;$$

here $1/A + 1/B \leq \max\{3/A, 2/B\} = d$. Thus $ed \leq \kappa$. The nonzero polynomial

$$F_0(v) = f_0(v(v+1), v^2(v+1))$$

has degree at most $W(\kappa - ed)$, since

$$2i + 3j \leq W(i/A + j/B)$$

for every monomial $x^i y^j$ of f_0 . Its expansions at $v = 0$ and $v = -1$ are both nonzero, since $k[v]$ injects into each completed local ring. These expansions are exactly the restrictions of f_0 to the two formal branches described above.

Let $\nu_t(f_0) = (\alpha, \beta)$ and $\mu = \alpha + t\beta$. Every term of its formal expansion has weight at least μ . Restricting to $\eta = 0$ therefore gives order at least μ on the branch at $v = 0$; restricting to $\xi = 0$ gives order at least μ/t on the branch at $v = -1$. The nonzero polynomial F_0 consequently satisfies

$$(1 + 1/t)\mu \leq \text{ord}_{v=0}(F_0) + \text{ord}_{v=-1}(F_0) \leq \deg(F_0) \leq W(\kappa - ed).$$

Equivalently, $(\alpha, \beta) \in (\kappa - ed)\Delta(a, b)$. Because $g = \xi\eta$, multiplicativity of least exponents gives

$$\nu_t(f) = (e, e) + \nu_t(f_0) = ed(c, c) + \nu_t(f_0) \in \kappa Q.$$

This also covers $\kappa = 0$, when f is constant. □

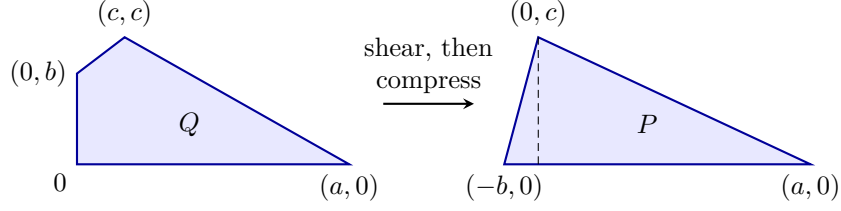


Figure 1: The nodal bound Q becomes the triangle P after a shear and vertical compression. Each vertical slice of the sheared bounding polygon is replaced by an interval of the same length starting at zero; Formula (4.3) gives the resulting boundary. Both polygons have area $H/2$. The drawing is schematic; the displayed vertex coordinates are exact.

4.2 A lattice direction of prescribed height

Choose $t > T$. With a, b as in Lemma 4.2, one has

$$a + b = W, \quad \frac{c}{a} + \frac{c}{b} = \frac{(1+t)^2}{Tt} > 1.$$

Thus Q is a quadrilateral with the vertices in the order displayed in that lemma, and its area is $c(a+b)/2 = H/2$. The shear $(\alpha, \beta) \mapsto (\alpha - \beta, \beta)$ sends its vertices to

$$(0,0), \quad (a,0), \quad (0,c), \quad (-b,b).$$

Its vertical slice at horizontal coordinate q has length

$$\begin{cases} c(1 + q/b), & -b \leq q \leq 0, \\ c(1 - q/a), & 0 \leq q \leq a. \end{cases} \quad (4.3)$$

Compression in the second coordinate therefore places the support in the triangle

$$P = \text{conv}\{(-b,0), (a,0), (0,c)\}. \quad (4.4)$$

Its area is again $H/2$; see Figure 1.

The next lemma chooses both t and a primitive lattice direction. We seek unimodular coordinates in which P has vertical extent H/w ; its area $H/2$ will then force its largest horizontal slice to have length w .

Lemma 4.3 (Prescribed primitive height). *If $w > 0$ and $dw < 1/16$, there exist $t > T$ and a primitive vector $u = (i, j) \in \mathbb{Z}^2$ such that the three vertex heights $z \mapsto \det(u, z)$ of the triangle P in (4.4) are distinct and their range has length H/w .*

Proof. Put $R = H/w$ and $a_0 = WT/(1+T)$. As t ranges over (T, ∞) , the value $a = Wt/(1+t)$ ranges over (a_0, W) . For $u = (i, j)$, the heights of $(-b,0), (a,0), (0,c)$ are $jb, -ja, i/d$. We seek $j > 0$ and i such that

$$ja + i/d = R, \quad i/d > jb.$$

The first equality turns the second inequality into $R - jW > 0$. Thus we first choose $jW \leq R/2$, and then use the freedom in a to obtain the equality with i coprime to j .

Take j to be the largest power of two at most

$$\frac{R}{2W} = \frac{1}{2dw}.$$

Then $j > 1/(4dw) > 4$. The open interval

$$(d(R - jW), d(R - ja_0))$$

has length

$$dj(W - a_0) = \frac{jT}{1 + T} > 2,$$

so it contains an odd integer i . Since j is a power of two, (i, j) is primitive. There is a unique $t > T$ such that

$$a = \frac{R - i/d}{j} \in (a_0, W).$$

The resulting heights satisfy

$$-ja < jb < i/d, \quad i/d - jb = R - jW \geq R/2 > 0, \quad i/d + ja = R.$$

This gives the required height range and distinctness. $\square$

4.3 Completion of the transformation

Proof of Proposition 4.1. Fix t and u from Lemma 4.3; these choices will be used in every degree. Keep coordinates $3, \dots, n$ unchanged. For each degree N and each label $\gamma = (\gamma_3, \dots, \gamma_n)$ occurring in S_N , put

$$\kappa = N - \sum_{h=3}^n \frac{\gamma_h}{a_h} \geq 0.$$

The pair exponents with this label lie in $\kappa\Delta(A, B)$. Take their polynomial span in $k[x, y]$ and replace it by its least exponents in the coordinates ξ, η and order ν_t . Lemma 2.2 preserves cardinality and gradedness: the row spaces multiply into the row with degree and label added, and the same coordinates and order are used throughout. For fixed N , choose a basis with distinct least exponents in each row and apply Lemma 2.3 to the finite union of these bases. Its common weight vector allows Lemma 2.4 to apply to all labels at once, giving backward jet transfer. The old points can be chosen in the dense open torus of $\mathbb{A}^2 \times \mathbb{T}^{n-2}$.

Lemma 4.2 places the new pair exponents in κQ . Apply the shear above and compress in the second coordinate. Lemmas 2.5 and 2.6, together with (4.3), give the bound κP . It remains to turn this triangle into $\Delta(w, H/w)$.

Complete the primitive vector u to an integral basis (u, v) with $\det(u, v) = 1$, and use the coordinates in that basis. This is a unimodular change, and its second coordinate is precisely $z \mapsto \det(u, z)$. Write $q_0 < q_1 < q_2$ for the three vertex heights of the transformed triangle. Its horizontal slice length is linear from zero at q_0 to a maximum at q_1 , and linear back to zero at q_2 . The maximum is w , since area is preserved and

$$\frac{H}{2} = \frac{1}{2}(q_2 - q_0) (\text{maximum slice length}), \quad q_2 - q_0 = R = H/w.$$

Compressing in the first coordinate therefore gives the bound

$$\text{conv}\{(0, q_0), (w, q_1), (0, q_2)\}.$$

Its vertical slice at first coordinate $h \in [0, w]$ has length $R(1 - h/w)$. Compression in the second coordinate gives $\Delta(w, R)$, as illustrated in Figure 2. These slice statements commute with multiplication by κ , so the resulting pair bound is $\kappa\Delta(w, R)$ for every label. If $\kappa = 0$, its only possible pair exponent is $(0, 0)$, which remains unchanged at every step.

Restoring the labels gives $S'_N \subset N\Delta(w, R, a_3, \dots, a_n)$. Every step preserves cardinality and gradedness and has backward jet transfer, so their composition has all the asserted properties. $\square$

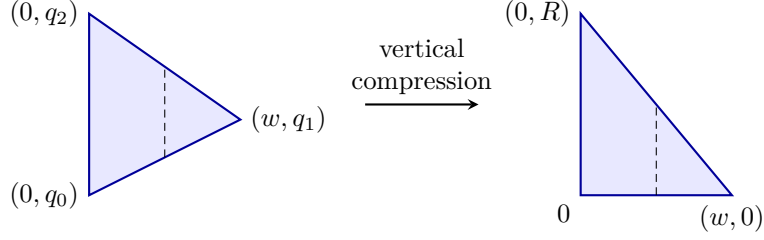


Figure 2: After the lattice change and horizontal compression, a final vertical compression produces the target triangle. At first coordinate h , both vertical slices have length $R(1 - h/w)$, where $R = q_2 - q_0 = H/w$. Dashed segments mark corresponding slices.

5 Interior filling and interpolation

The reshaping construction preserves the number of exponents, but does not identify the individual exponents that survive. Gradedness supplies the missing information: a family with full asymptotic density contains every lattice point in a fixed compact subset of the interior, in all sufficiently large degrees. We first prove this statement, then use it to find a complete interpolation space inside the reshaped supports. These two arguments follow [11, Lemmas 5.1 and 5.2]. Interior filling for semigroups is studied more generally in [5, Theorem 1.6]; the full-density hypothesis here permits a direct proof by counting decompositions.

Lemma 5.1 (Interior filling). *Let $P \subset \mathbb{R}^n$, $n \geq 1$, be a compact convex set with nonempty interior. Suppose finite sets $F_N \subset NP \cap \mathbb{Z}^n$, indexed by integers $N \geq 1$, satisfy*

$$F_N + F_{N'} \subset F_{N+N'} \quad \text{and} \quad \#F_N = \text{vol}(P)N^n + o(N^n).$$

Then for every compact set $K \subset \text{int}(P)$ there is an integer N_K such that

$$NK \cap \mathbb{Z}^n \subset F_N \quad (N \geq N_K).$$

Proof. The boundary of a convex body has measure zero, so lattice-point counting by Riemann sums gives

$$\#(NP \cap \mathbb{Z}^n) = \text{vol}(P)N^n + o(N^n).$$

Consequently the number of missing lattice points

$$b_N := \#((NP \cap \mathbb{Z}^n) \setminus F_N)$$

is $o(N^n)$. We show that a point of NK has too many decompositions into two points in nearly equal degrees for all of them to involve a missing point.

Assume K is nonempty, and choose $\rho > 0$ so that the closed ball of radius ρ about every point of K lies in P . Write $N = N_1 + N_2$ with $N_1 = \lfloor N/2 \rfloor$ and $N_2 = \lceil N/2 \rceil$. For $y \in NK \cap \mathbb{Z}^n$, consider all lattice points

$$y_1 \in \mathbb{Z}^n, \quad \left\| y_1 - \frac{N_1}{N} y \right\| < \rho N_1, \quad y_2 := y - y_1.$$

Both y_1/N_1 and y_2/N_2 lie in the radius- ρ ball about y/N . Thus $y_i \in N_i P \cap \mathbb{Z}^n$. The ball from which y_1 is chosen contains at least cN^n lattice points for a constant $c > 0$ independent of y and all sufficiently large N : an inscribed cube gives this bound uniformly in the center. Among these choices, at most b_{N_1} have $y_1 \notin F_{N_1}$, and at most b_{N_2} have $y_2 \notin F_{N_2}$, since $y_1 \mapsto y - y_1$ is injective. As $b_{N_1} + b_{N_2} = o(N^n)$, at least one choice has $y_i \in F_{N_i}$. Gradedness then gives $y = y_1 + y_2 \in F_N$. All bounds are uniform in y , which proves the assertion. $\square$

The interpolation space we need is a simplex stretched in the last coordinate by the number of points. Its fibers over the other exponents have exactly the lengths needed for one-variable interpolation with multiplicities.

Lemma 5.2 (Jets from an elongated simplex). *Let $n, r \geq 1$ and $m \geq 0$ be integers, and let k be a field containing distinct nonzero elements $c_1, \dots, c_r$. Set*

$$E_m = \left\{ (\beta, b) \in \mathbb{Z}_{\geq 0}^{n-1} \times \mathbb{Z}_{\geq 0} : |\beta| + \frac{b}{r} < m + 1 \right\}, \quad |\beta| = \sum_{j=1}^{n-1} \beta_j. \quad (5.1)$$

Then $M(E_m)$ separates jets through order m at the r points

$$q_i = (1, \dots, 1, c_i) \in \mathbb{T}^n(k).$$

For $n = 1$, the tuple β is empty and $|\beta| = 0$.

Proof. Write the coordinates as $(x_1, \dots, x_{n-1}, Y)$ and put $x' - 1 = (x_1 - 1, \dots, x_{n-1} - 1)$. An arbitrary jet through order m at q_i has a unique expression

$$\sum_{|\alpha| \leq m} (x' - 1)^\alpha h_{i,\alpha}(Y - c_i), \quad \deg h_{i,\alpha} \leq m - |\alpha|.$$

For each α , the Chinese remainder theorem gives a polynomial $h_\alpha(Y)$ satisfying

$$h_\alpha(Y) \equiv h_{i,\alpha}(Y - c_i) \pmod{(Y - c_i)^{m-|\alpha|+1}} \quad (1 \leq i \leq r), \quad \deg h_\alpha < r(m - |\alpha| + 1).$$

Indeed, the ideals in this display are pairwise comaximal, and their product is generated by a monic polynomial of degree $r(m - |\alpha| + 1)$. The polynomial

$$\sum_{|\alpha| \leq m} (x' - 1)^\alpha h_\alpha(Y)$$

realizes the prescribed jets. Each monomial $x'^\beta Y^b$ in its expansion satisfies $\beta_j \leq \alpha_j$ for every j and $b < r(m - |\alpha| + 1)$ for some α . Hence $|\beta| + b/r < m + 1$, so the polynomial belongs to $M(E_m)$. This uses powers of local parameters and the Chinese remainder theorem, and therefore works in every characteristic. $\square$

We now choose the number of points large enough that successive applications of Proposition 4.1 produce the required elongated simplex.

Proposition 5.3 (Jets at the geometric generic tuple). *Let k be an algebraically closed field of positive characteristic, let X/k be a smooth integral projective variety of dimension $n \geq 2$, and let L be ample. Put $V = L^n$. There is an integer $r_0 = r_0(X, L)$ such that, for every integer $r \geq r_0$ and every real number*

$$0 < \theta < \left(\frac{V}{r} \right)^{1/n},$$

there is an integer $N_0 = N_0(r, \theta)$ such that, for every $N \geq N_0$, the sections of NL separate jets through order $\lfloor N\theta \rfloor$ at some ordered r -tuple of distinct k -points, which may depend on N . For each such N , the same jet separation holds after base change to K_r at the geometric generic tuple $p_1, \dots, p_r$.

Proof. Take the graded family S_N and simplex $P_0 = \Delta(a_1, \dots, a_n)$ from Proposition 3.1. Thus $\prod_j a_j = V$, the support S_N lies in NP_0 , and

$$\#S_N = \frac{V}{n!}N^n + o(N^n).$$

Let $\ell = \min_j a_j$, and choose r_0 so large that

$$w := \left(\frac{V}{r}\right)^{1/n} < \frac{\ell}{100} \quad (r \geq r_0).$$

Fix one such r . Reshape the first two intercepts to $(w, a_1 a_2 / w)$, then reshape the second and third intercepts, and continue through the $(n-1)$ st and n th intercepts. At every step, both intercepts A, B being reshaped are at least ℓ : untouched intercepts have this property, and the carried intercept AB/w is larger than A because $B > w$. Consequently

$$w \max(3/A, 2/B) \leq \frac{3w}{\ell} < \frac{1}{16},$$

as required by Proposition 4.1. The product of all intercepts remains V . The resulting graded family F_N therefore satisfies

$$\begin{aligned} F_N &\subset NP_* \cap \mathbb{Z}^n, & P_* &:= \Delta(w, \dots, w, rw), \\ \#F_N &= \frac{V}{n!}N^n + o(N^n) = \text{vol}(P_*)N^n + o(N^n). \end{aligned} \tag{5.2}$$

Here the last intercept is $V/w^{n-1} = rw$. The transformations preserve backward jet transfer, so jet separation by $M(F_N)$ will imply jet separation by $H^0(X, NL)$.

Fix $0 < \theta < w$. To apply Lemma 5.1, we contract P_* and then translate it away from the coordinate faces. Choose λ with $\theta/w < \lambda < 1$, and choose a vector $v \in \mathbb{R}_{>0}^n$ small enough that

$$v + \lambda P_* \subset \text{int}(P_*).$$

For example, it suffices that $\sum_{j < n} v_j/w + v_n/(rw) < 1 - \lambda$. Choose vectors $v_N \in N^{-1}\mathbb{Z}^n$ tending to v . For all sufficiently large N , the sets $v_N + \lambda P_*$ lie in a common compact subset K of $\text{int}(P_*)$. Applying Lemma 5.1 to (5.2) gives

$$Nv_N + (N\lambda P_* \cap \mathbb{Z}^n) \subset F_N \tag{5.3}$$

for all sufficiently large N .

Put $m = \lfloor N\theta \rfloor$. Since $\theta < \lambda w$, we have $m+1 \leq N\lambda w$ for all sufficiently large N . The defining inequality (5.1) therefore gives

$$E_m \subset N\lambda P_* \cap \mathbb{Z}^n.$$

Choose distinct $c_1, \dots, c_r \in k^\times$ and apply Lemma 5.2. Multiplication by the Laurent monomial with exponent $Nv_N \in \mathbb{Z}^n$ is multiplication by a unit on the torus, hence induces an automorphism on every jet space. Thus (5.3) implies that $M(F_N)$ separates jets through order m at these r points. Backward jet transfer through all reshapings and Proposition 3.1 gives the asserted jet separation at an ordered tuple on X .

For each fixed r, θ , and sufficiently large N , the locus in U_r where this evaluation map is surjective is a nonempty open set, by the openness of jet separation established in Section 2. The variety U_r is integral, so its generic point belongs to that open set. Base change of the jet evaluation map to K_r preserves surjectivity and identifies its targets with the jets at $p_1, \dots, p_r$. The same geometric generic tuple therefore satisfies every required jet-separation statement. $\square$

6 From jets to the Seshadri constant

The interpolation statement now gives the lower bound for the Seshadri constant. We first spell out the passage from jets to intersection numbers, including at singular points of a curve, and then recall the matching volume bound. The relation between asymptotic jet separation and Seshadri constants is classical; see [2, Theorem 6.4] in the complex setting and [8, Section 2, Equation (2.3)] in arbitrary characteristic. The arguments below work in arbitrary characteristic.

Lemma 6.1 (Jets and curve multiplicities). *Let Y be a smooth integral projective variety over an algebraically closed field, let A be a line bundle on Y , and let $q_1, \dots, q_r$ be distinct points. Suppose that $H^0(Y, A)$ separates jets through order $m \geq 1$ at these points. Then every integral curve $C \subset Y$ meeting at least one of them satisfies*

$$A \cdot C \geq m \sum_{i=1}^r \text{mult}_{q_i} C,$$

where the multiplicity is zero at a point outside C .

Proof. Choose such a point q_j . Write $R = \mathcal{O}_{C, q_j}$ and let $\mathfrak{n}$ be its maximal ideal. The vector space $\mathfrak{n}^m / \mathfrak{n}^{m+1}$ is nonzero: otherwise Nakayama's lemma would give $\mathfrak{n}^m = 0$, contrary to R being a one-dimensional local domain. In a local frame of A , lift a nonzero class in this vector space to an ambient jet of order m at q_j . This lift can be chosen in the image of $\mathfrak{m}_{Y, q_j}^m$. Prescribe that jet at q_j and the zero jet at all other marked points. Jet separation produces a section $\sigma \in H^0(Y, A)$ which vanishes to order at least m at every q_i and whose restriction to C is nonzero.

For completeness, the local multiplicity estimate used here is

$$\text{length}_R(R/(f)) \geq m e(R) \quad (0 \neq f \in \mathfrak{n}^m), \quad (6.1)$$

where $e(R)$ denotes the Hilbert–Samuel multiplicity of a one-dimensional Noetherian local domain $(R, \mathfrak{n})$; see [15, Definition 43.15.1, Tag 0AZV]. Indeed, for positive integers t , the nonzero divisor f gives

$$t \text{length}_R(R/(f)) = \text{length}_R(R/(f^t)) \geq \text{length}_R(R/\mathfrak{n}^{mt}) = mt e(R) + O(1).$$

Divide by t and let t tend to infinity. Applying (6.1) to the local equations of $\sigma|_C$ at the marked points proves the inequality: the sum of all local zero lengths of a nonzero section of $A|_C$ is $\deg(A|_C) = A \cdot C$. $\square$

Lemma 6.2 (Volume bound). *Let Y be a smooth integral projective variety of dimension $d \geq 2$ over an algebraically closed field, let A be ample, and let $q_1, \dots, q_r$ be distinct points. Then*

$$\inf_{C \cap \{q_1, \dots, q_r\} \neq \emptyset} \frac{A \cdot C}{\sum_i \text{mult}_{q_i} C} \leq \left(\frac{A^d}{r} \right)^{1/d},$$

where the infimum is over integral curves in Y .

Proof. Denote the infimum by ε . If $\varepsilon = 0$, the conclusion is immediate. Otherwise, blow up the marked points:

$$\pi : \tilde{Y} \longrightarrow Y, \quad E_1, \dots, E_r \subset \tilde{Y}.$$

For each rational number $0 \leq s < \varepsilon$, the divisor class

$$D_s = \pi^* A - s \sum_i E_i$$

is nef. To check this on curves, first observe that $D_s|_{E_i} = s\mathcal{O}_{\mathbb{P}^{d-1}}(1)$, which has nonnegative degree on every curve in E_i . Every other integral curve in $\tilde{Y}$ is the strict transform $\tilde{C}$ of an integral curve $C \subset Y$, and

$$E_i \cdot \tilde{C} = \text{mult}_{q_i} C. \quad (6.2)$$

Thus $D_s \cdot \tilde{C} \geq 0$ by the definition of ε ; for curves missing all marked points this follows from the ampleness of A .

Here is a local justification of (6.2), which also covers singular curves. If $q_i \in C$, the Rees algebra construction identifies $\tilde{C}$ with the blow-up of C at the marked points. Indeed, the ambient Rees algebra surjects onto the Rees algebra of the restricted ideal. The latter is integral, and its Proj agrees with C away from the centers, so its closed image is exactly the strict transform. For $R = \mathcal{O}_{C, q_i}$ with maximal ideal $\mathfrak{n}$, its exceptional fiber is

$$\text{Proj}(\text{gr}_{\mathfrak{n}} R).$$

It is the effective Cartier divisor cut out on $\tilde{C}$ by E_i . Its length is $e(R)$: the Hilbert function $\dim_{k(q_i)}(\mathfrak{n}^t/\mathfrak{n}^{t+1})$ equals $e(R)$ for all sufficiently large t , and its eventual value is the length of this zero-dimensional projective scheme. This proves (6.2); if $q_i \notin C$, both sides vanish.

A nef class on a projective variety is a limit of ample classes and therefore has nonnegative top self-intersection. The exceptional divisors are disjoint, π^*A restricts trivially to each of them, and $E_i^d = (-1)^{d-1}$. Consequently

$$0 \leq D_s^d = A^d - rs^d.$$

If $\varepsilon > (A^d/r)^{1/d}$, even if $\varepsilon = +\infty$, there is a rational number s strictly between $(A^d/r)^{1/d}$ and ε . The displayed inequality excludes such an s , proving the bound. $\square$

Proof of Theorem 1.1. Choose r_0 as in Proposition 5.3 and fix $r \geq r_0$. Put $V = L^n$ and $w = (V/r)^{1/n}$. Work over K_r at the geometric generic tuple $p_1, \dots, p_r$. For every fixed real number $0 < \theta < w$ and all sufficiently large integers N , Proposition 5.3 gives separation of jets through order $m = \lfloor N\theta \rfloor$ by $H^0(X_{K_r}, L_{K_r}^{\otimes N})$ at this fixed tuple.

For large N we have $m \geq 1$, so Lemma 6.1 yields

$$\frac{L_{K_r} \cdot C}{\sum_i \text{mult}_{p_i} C} \geq \frac{\lfloor N\theta \rfloor}{N}$$

for every integral curve in the defining infimum. Letting N tend to infinity gives $\varepsilon_r(X, L) \geq \theta$. Since $\theta < w$ was arbitrary, $\varepsilon_r(X, L) \geq w$. Intersection numbers are unchanged by extension of the algebraically closed ground field, so $(L_{K_r})^n = V$; Lemma 6.2 supplies the reverse inequality. $\square$

Remark 6.3 (Surfaces). The same proof gives the conclusion of Theorem 1.1 when $n = 2$. Indeed, Propositions 3.1, 4.1, and 5.3 all allow $n = 2$; the reshaping is then performed just once, with no unchanged coordinates. Both intersection lemmas above also apply in dimension two.

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